Per-subproblem Benders cuts and the CVaR link

Stochastic_CapExpansion

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This note compares the master problem as it is built today, with one optimality cut per scenario, against the proposed version, with one cut per (scenario, time block) subproblem and the CVaR auxiliary variables tied to the block cost variables. It shows that the proposed master gives a valid lower bound, never a looser one, and that for one block per scenario it is the same problem as today's with fewer rows.

1 Setting and notation

- $x \in X$: first-stage decisions (resource capacities and line expansions) with investment cost $c^\top x$. X is the box and fixed-capacity set the master already imposes.
- $\omega \in \mathcal{W}$: operational scenarios $\omega = (s, f, k)$ (demand, gas, weather) with probabilities $p_\omega > 0$, $\sum_\omega p_\omega = 1$.
- $b = 1, \dots, B$: time blocks of a scenario ("Subproblems per scenario"). Each block is a run of whole representative periods, and no constraint links two periods.
- $\theta_{\omega,b}(x) \geq 0$: optimal operating cost of block b of scenario ω at capacity x. It is the value of an LP whose right-hand side is affine in x, so it is convex and piecewise linear in x. Because blocks don't interact, the scenario cost is

$$\theta_\omega(x) = \sum_{b=1}^{B} \theta_{\omega,b}(x). \tag{1}$$

- $\Omega \in [0, 1]$: risk weight (the code's `risk_aversion_weight`), and $\Psi \in (0, 1]$: tail fraction (`Value-at-Risk percent`).
- $\mathcal{J}$: Benders iterates solved so far. Iterate j solves every block at x^j and returns $\theta_{\omega,b}(x^j)$ and a subgradient $\beta^j_{\omega,b}$ (the capacity duals). The block tangent is

$$\ell^j_{\omega,b}(x) = \theta_{\omega,b}(x^j) + \beta^{j\top}_{\omega,b}\,(x - x^j) \;\leq\; \theta_{\omega,b}(x) \qquad \forall x, \tag{2}$$

  by convexity. Rows from a subproblem that did not solve to optimality have a zero coefficient on the cost variable and are ignored throughout.

The true risk-adjusted objective is

$$F(x) = c^\top x + \Omega \sum_\omega p_\omega\, \theta_\omega(x) + (1 - \Omega)\ \mathrm{CVaR}_\Psi\big(\theta(x)\big), \tag{3}$$

with the Rockafellar–Uryasev form

$$\mathrm{CVaR}_{\Psi}(v) = \min_{\zeta \in \mathbb{R}} \Big\{ \zeta + \tfrac{1}{\Psi} \sum_{\omega} p_\omega \max(0,\, v_\omega - \zeta) \Big\}. \tag{4}$$

CVaR_{Ψ} is nondecreasing in each v_ω. The risk-neutral case is $\Omega = 1$ with no u, ζ variables.

2 The master problem today

Today the block results are summed before the cut is built, so each iterate gives one cut per scenario, $\sum_b \ell^j_{\omega,b}$:

$$\begin{aligned}
(\mathrm{M_{agg}}) \quad \min_{x \in X,\ \alpha,\ u,\ \zeta} \quad & c^\top x + \Omega \sum_{\omega} p_\omega\, \alpha_\omega + (1-\Omega)\Big[\zeta + \tfrac{1}{\Psi} \sum_{\omega} p_\omega u_\omega\Big] & & \\
\text{s.t.} \quad & \alpha_\omega \ge \textstyle\sum_b \ell^j_{\omega,b}(x) & \forall \omega,\ j \in \mathcal{J} \quad & (5)\\
& u_\omega \ge \textstyle\sum_b \ell^j_{\omega,b}(x) - \zeta & \forall \omega,\ j \in \mathcal{J} \quad & (6)\\
& u_\omega \ge -\zeta & \forall \omega \quad & (7)\\
& \alpha_\omega \ge 0,\ u_\omega \ge 0. &&
\end{aligned}$$

In the code, (5) are the `optimality_cut_*` rows, (6) the `cvar_tail_cuts_*` rows (the same tangent again, minus ζ), and (7) the fixed `cvar_tail` rows. Write the aggregated scenario estimate as

$$\widetilde{L}_\omega(x) = \max\Big(0,\ \max_{j \in \mathcal{J}} \sum_b \ell^j_{\omega,b}(x)\Big). \tag{8}$$

The row $u_\omega \ge -\zeta$ is the CVaR cut for the constant tangent 0, which is valid because costs are nonnegative. So the three u rows together say $u_\omega \ge \widetilde{L}_\omega(x) - \zeta$.

3 The proposed master problem

Each block gets its own cost variable and its own cuts, and the CVaR variables are tied to the sum of the block variables by one fixed row per scenario:

$$\begin{aligned}
(\mathrm{M_{blk}}) \quad \min_{x \in X,\ \alpha,\ u,\ \zeta} \quad & c^\top x + \Omega \sum_{\omega} p_\omega \sum_b \alpha_{\omega,b} + (1-\Omega)\Big[\zeta + \tfrac{1}{\Psi} \sum_{\omega} p_\omega u_\omega\Big] & & \\
\text{s.t.} \quad & \alpha_{\omega,b} \ge \ell^j_{\omega,b}(x) & \forall \omega,\ b,\ j \in \mathcal{J} \quad & (9)\\
& u_\omega \ge \textstyle\sum_b \alpha_{\omega,b} - \zeta & \forall \omega \quad & (10)\\
& \alpha_{\omega,b} \ge 0,\ u_\omega \ge 0. &&
\end{aligned}$$

Here (9) are the `optimality_cut_*` rows, now one per block, and (10) replaces the fixed `cvar_tail` rows. It is built once, with the master, and no CVaR cut rows are added during the iterations. The block estimate is

$$\widehat{L}_\omega(x) = \sum_{b=1}^{B} \max\Big(0,\ \max_{j \in \mathcal{J}} \ell^j_{\omega,b}(x)\Big). \tag{11}$$

In code, $\sum_b \alpha_{\omega,b}$ is the expression `scenario_alpha`, and `expected_alpha` and `cvar_term` keep their current meaning.

4 Results

For a fixed x, let $V_{\mathrm{agg}}(x)$ and $V_{\mathrm{blk}}(x)$ be the optimal values of the two masters over the remaining variables (α, u, ζ).

Proposition 1 (Reduced objectives)**.** *For every $x \in X$ (and $\Omega < 1$; for $\Omega = 1$ drop the CVaR term),*

$$V_{agg}(x) = c^\top x + \Omega \sum_\omega p_\omega \widetilde{L}_\omega(x) + (1-\Omega)\ \mathrm{CVaR}_\Psi\left(\widetilde{L}(x)\right), \tag{12}$$

$$V_{blk}(x) = c^\top x + \Omega \sum_\omega p_\omega \widehat{L}_\omega(x) + (1-\Omega)\ \mathrm{CVaR}_\Psi\left(\widehat{L}(x)\right). \tag{13}$$

Proof. Take ($\mathrm{M_{blk}}$) with x fixed. The rows involving $\alpha_{\omega,b}$ are its cuts, $\alpha_{\omega,b} \geq 0$ and the link (10). The objective coefficient of $\alpha_{\omega,b}$ is $\Omega p_\omega \geq 0$, and in (10) a smaller $\alpha_{\omega,b}$ only relaxes the bound on u_ω. So, for any feasible point, replacing each $\alpha_{\omega,b}$ by its smallest feasible value $a_{\omega,b} = \max(0, \max_j \ell^j_{\omega,b}(x))$ keeps it feasible and does not raise the objective. We may therefore set $\alpha_{\omega,b} = a_{\omega,b}$, so $\sum_b \alpha_{\omega,b} = \widehat{L}_\omega(x)$.

With that, the smallest feasible u_ω for a given ζ is $\max(0, \widehat{L}_\omega(x) - \zeta)$, and u_ω has the nonnegative coefficient $(1-\Omega)p_\omega/\Psi$, so it takes that value. What remains is

$$c^\top x + \Omega \sum_\omega p_\omega \widehat{L}_\omega(x) + (1-\Omega) \min_\zeta \left\{ \zeta + \tfrac{1}{\Psi} \sum_\omega p_\omega \max\left(0, \widehat{L}_\omega(x) - \zeta\right) \right\},$$

and the minimum over ζ is $\mathrm{CVaR}_\Psi(\widehat{L}(x))$.

($\mathrm{M_{agg}}$) is the same argument with one block: the smallest feasible α_ω is $\widetilde{L}_\omega(x)$, and the three families of u rows give $u_\omega \geq \max\left(0, \max(0, \max_j \sum_b \ell^j_{\omega,b}(x)) - \zeta\right) = \max(0, \widetilde{L}_\omega(x) - \zeta)$. □

Proposition 2 (Valid and never looser)**.** *For every x and every scenario ω,*

$$\widetilde{L}_\omega(x) \ \leq\ \widehat{L}_\omega(x) \ \leq\ \theta_\omega(x). \tag{14}$$

Hence $V_{agg}(x) \leq V_{blk}(x) \leq F(x)$ for every x, and the optimal value of (M_{blk}) is a lower bound on $\min_x F(x)$ that is at least as tight as today's.

Proof. Upper inequality. Each block term satisfies $\max(0, \max_j \ell^j_{\omega,b}(x)) \leq \theta_{\omega,b}(x)$, because every tangent is below $\theta_{\omega,b}$ and $\theta_{\omega,b} \geq 0$. Summing over b gives $\widehat{L}_\omega(x) \leq \theta_\omega(x)$.

Lower inequality. For any iterate j and each block, $\ell^j_{\omega,b}(x) \leq \max(0, \max_{j'} \ell^{j'}_{\omega,b}(x))$, so summing over b, $\sum_b \ell^j_{\omega,b}(x) \leq \widehat{L}_\omega(x)$ for every j. Also $\widehat{L}_\omega(x) \geq 0$. Taking the maximum over j and over 0 gives $\widetilde{L}_\omega(x) \leq \widehat{L}_\omega(x)$.

The bounds on V follow from Proposition 1: both the expectation and CVaR_Ψ are nondecreasing in the scenario values, and $\Omega, 1-\Omega \geq 0$. □

The gain is that each block may take its best tangent from a different iterate. For example, a summer week can use the tangent from iterate 3 and a winter week the one from iterate 7. Today one iterate has to supply all of a scenario's blocks at once. The lower inequality is strict whenever no single iterate is best for every block of a scenario at x.

Corollary 1 (One block per scenario)**.** *If $B = 1$, then $\widehat{L}_\omega = \widetilde{L}_\omega$ for every ω. So $V_{blk} = V_{agg}$, and (M_{blk}) has the same optimal x, lower bound and CVaR term as today's master. It replaces the per-iteration* `cvar_tail_cuts` *rows with the single fixed row* (10) *per scenario.*

Proof. With $B = 1$, $\widehat{L}_\omega(x) = \max(0, \max_j \ell^j_{\omega,1}(x)) = \widetilde{L}_\omega(x)$. Since the two reduced objectives in Proposition 1 are then the same function of x, the two masters have the same minimizers and optimal value. □

Remark (Why tie CVaR to the block variables instead of keeping CVaR cuts). CVaR cuts on u_ω can only use one iterate's tangent per row: $u_\omega \geq \sum_b \ell^j_{\omega,b}(x) - \zeta$. That bounds u_ω by $\widetilde{L}_\omega(x) - \zeta$ however many block cuts there are, so the CVaR term would stay at today's looser estimate while the expectation term gets $\widehat{L}$. With the link (10), the CVaR term picks up $\widehat{L}$ too (Proposition 1), and the master has fewer rows.

Remark ($\Omega = 0$, which is the same as today). When $\Omega = 0$, α appears in the objective only through u. For a scenario outside the tail, the link (10) is slack, so $\alpha_{\omega,b}$ can take any value at or above its cuts with the same objective. The optimal x, the lower bound $c^\top x + \mathrm{CVaR}_\Psi(\widehat{L}(x))$ and the feasible set in x are unaffected, since the argument of Proposition 1 still gives the optimal value. Only the reported `Alpha` values can sit above $\widehat{L}$. Today, at $\Omega = 0$, α has zero objective weight and appears only in its cuts (and in any budget rows), so it is at least as loose. Budget constraints that contain `expected_alpha` only become harder to meet if α rises, so they do not change which x is feasible either.

5 What else stays the same

- **Lower bound** reported by the Benders loop: $\Omega \cdot \texttt{expected_alpha} + (1 - \Omega) \cdot \texttt{cvar_term} + c^\top x$, unchanged in form. By Proposition 1 it equals V_{blk} at the master solution.
- **Upper bound**: computed from the true scenario costs $\theta_\omega(x^j)$, as now.
- **GridFill and the cut export** use cuts without the master, so they get the same block form. A scenario's lower bound at a blend $\bar{x}$ becomes $\mathrm{LB}_\omega(\bar{x}) = \sum_b \max_{k \in (\omega,b)}(a_k + g_k^\top \bar{x})$ instead of $\max_{k \in \omega}(a_k + g_k^\top \bar{x})$. It is valid for the same reason as the upper inequality of Proposition 2.

6 Size of the master

Per Benders iteration, with $|\mathcal{W}|$ scenarios and B blocks per scenario:

	Today ($\mathrm{M}_{\mathrm{agg}}$)	Proposed ($\mathrm{M}_{\mathrm{blk}}$)
Cost variables	$\lvert\mathcal{W}\rvert$	$B\,\lvert\mathcal{W}\rvert$
Optimality cut rows / iteration	$\lvert\mathcal{W}\rvert$	$B\,\lvert\mathcal{W}\rvert$
CVaR cut rows / iteration	$\lvert\mathcal{W}\rvert$ (risk on)	0
Fixed CVaR rows	$\lvert\mathcal{W}\rvert$ ($u \geq -\zeta$)	$\lvert\mathcal{W}\rvert$ (link)

With risk on, $B = 1$ halves the cut rows added per iteration. With $B = 12$ (one per week on the 16-zone case) each iteration adds $12|\mathcal{W}|$ rows, six times today's $2|\mathcal{W}|$. The cut pool's pruning and archiving keep the live set small. Fewer iterations should offset part of the growth, but the "Cuts retained" cap may need raising for large runs.